

Citizen Grievance & Sentiment Analysis System

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Objective

The main objective of this week was to deploy the trained machine learning model using FastAPI and build a real-time complaint analysis system that can process user input and return predictions through an API.

Task Completed

During Week 4, the focus was on transforming the machine learning model into a usable application. The following tasks were completed:

1. Model Integration

- Loaded the trained machine learning model (model.pkl) and TF-IDF vectorizer (vectorizer.pkl) using pickle.
- Ensured the model can handle real-time input.

2. API Development using FastAPI

- a. Created an API using FastAPI framework.
- b. Defined endpoints such as:
 - i. / → to check API status
 - ii. /predict → to analyze complaints

3. Text Processing Implementation

- a. Applied text cleaning techniques such as:
 - i. Lowercasing
 - ii. Removing special characters
 - iii. Stopword removal

4. Prediction Logic

- a. Implemented a function to:
 - i. Predict complaint type
 - ii. Analyze sentiment using TextBlob
 - iii. Assign priority (High, Medium, Low)

5. Testing using Swagger UI

- Tested API endpoints using /docs interface.
- Verified real-time predictions for multiple complaint inputs.

Tools & Technologies Used

1. Python
2. FastAPI
3. Scikit-learn
4. TF-IDF Vectorizer
5. TextBlob (Sentiment Analysis)
6. Uvicorn (ASGI Server)
7. Jupyter Notebook
8. Anaconda

Outcome of the Week

- Successfully deployed a machine learning model as an API.
- Built a system that takes user complaints and returns:
 - Complaint Type
 - Sentiment
 - Priority Level
- Enabled real-time interaction with the model.

Challenges Faced

- Error in loading FastAPI application due to incorrect file structure.
- Difficulty in locating correct project directory in terminal.
- Handling special characters (like &) in folder paths.
- Understanding separation between model training and API deployment.

Learning Outcomes

- Learned how to deploy ML models using FastAPI.
- Gained knowledge of API development and routing.
- Understood how backend systems handle real-time data.
- Improved debugging and problem-solving skills.

Conclusion

Week 4 focused on deployment and integration of the machine learning model into a real-world application. The successful implementation of FastAPI enabled real-time complaint analysis, making the system more practical and user-friendly.